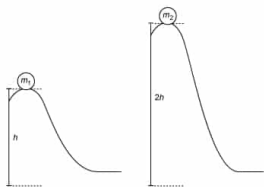


Two identical balls of equal mass m_1 and m_2 are placed at rest at the top of separate hills.

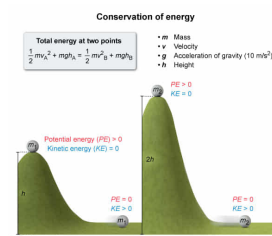


How do the velocities v_1 and v_2 of the balls compare, measured after each has rolled down to the bottom of its respective hill? (Note: Assume the absence of friction and negligible rotational kinetic energy.)

- ☐ A. $v_1 = \frac{1}{2} \cdot v_2$ [35%]
- ☒ B. $v_1 = \frac{\sqrt{2}}{2} \cdot v_2$ [45%]
- ☐ C. $v_1 = 2 \cdot v_2$ [13%]
- ☐ D. $v_1 = 4 \cdot v_2$ [5%]

Correct
45% answered correctly

Explanation:



In mechanical systems, the total energy (U) of an object is quantified by the sum of kinetic energy (KE) and potential energy (PE):

$$U = KE + PE$$

KE describes the energy of translational or rotational motion. Specifically, [translational kinetic energy](#) refers to the energy associated with the motion of an object through three-dimensional space and relates to the mass (m) and velocity (v) of an object:

$$KE = \frac{1}{2} m v^2$$

PE describes the stored energy associated with an object's configuration (eg, a compressed spring) or relative position. [Gravitational potential energy](#) describes the energy an object possesses when placed at some distance above the Earth's surface, and it relates to the object mass (m), gravitational constant ($g = 10 \text{ m/s}^2$), and height (h):

$$PE = m g h$$

Conservation of energy requires that an object's **total energy** observed between two points (A and B) **remains the same**, but contributions from potential energy may convert to kinetic energy (or vice versa) within mechanical systems:

$$\frac{1}{2} m v_A^2 + m g h_A = \frac{1}{2} m v_B^2 + m g h_B$$

In this question, each of two balls of equal mass (m_1 and m_2) is at rest at the top of a hill, each with a different height (h and $2h$). Based on the conservation of energy, all the potential energy stored within the balls at the top of the hills is converted into kinetic energy as the balls reach the bottom of the hills (ie, the earth's surface) such that:

$$m_1 g h = \frac{1}{2} m_1 v_1^2 \quad m_2 g (2h) = \frac{1}{2} m_2 v_2^2$$

The velocity of each ball may be determined by rearranging and canceling terms within the equations above:

$$v_1 = \sqrt{\frac{2 g h}{1}} \quad v_2 = \sqrt{\frac{2 g (2h)}{1}} = \sqrt{4 g h} = 2 \sqrt{g h}$$

Consequently, the ratio between the velocity of each ball is:

$$\frac{v_1}{v_2} = \frac{\sqrt{2 g h}}{2 \sqrt{g h}} = \frac{\sqrt{2}}{2} \approx .70$$

Therefore, placing a ball on a hill at half the height of an equal ball results in the first ball having 70% of the velocity of the second ball upon reaching the bottom of the hill (**Choice B**).

Educational objective:

Translational kinetic energy is the energy of motion whereas gravitational potential energy describes the energy of position. Due to the conservation of energy, in the absence of friction, the total energy of an object along a path remains constant, but contributions from potential energy may convert to kinetic energy (or vice versa).

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